



KNOWLEDGE SYSTEMS WORKING GROUP

Bridging Knowledge & AI Across Engineering Disciplines

Enabling semantic interoperability through vocabularies,
taxonomies, ontologies, and prompt engineering

————— INCOSE IW 2026 —————

THE PROBLEM

50+ Working Groups. One Language Barrier.



Incompatible Terminologies

"Requirement" means different things in Requirements WG, MBSE Initiative, and Healthcare WG



Redundant Development

Multiple WGs independently build overlapping frameworks without coordination



AI Cannot Leverage Our Knowledge

Without semantic grounding, AI systems hallucinate on SE topics



Missed Cross-Domain Insights

No infrastructure exists to synthesize expertise across disciplines

What's at Stake for INCOSE



Lost Innovation

Critical cross-domain insights require multiple WGs but no infrastructure enables this collaboration



AI Readiness Gap

Organizations leveraging AI-ready knowledge structures are outpacing those with fragmented data



Diminished Impact

Siloed working groups deliver less value than a connected ecosystem could achieve

The question isn't whether to act—it's how quickly we can enable interoperability

Every month of delay compounds fragmentation and missed opportunities

Knowledge Engineering + Prompt Engineering



Knowledge Engineering

Vocabularies — Standardized terms across all WGs

Taxonomies — Hierarchical classification systems

Ontologies — Formal semantic relationships



Prompt Engineering

RAG 2.0 — Knowledge graph-enhanced AI responses

Semantic Guardrails — AI outputs aligned to standards

Human-AI Patterns — Explainable, trustworthy AI



Unified Knowledge Fabric enabling AI-ready collaboration

What Your Working Group Gains



Understanding

Shared conceptual models let WGs understand each other's domains without translation



Coordination

Enterprise knowledge graphs provide infrastructure for cross-group initiatives



Communication

Unified semantic layer enables unambiguous communication across all 50+ WGs



Impact

AI-enhanced decision-making with 40% improved accuracy through RAG 2.0

50+

WGs Connected

4.8x

First-Year ROI

40%

AI Accuracy Gain

1

Unified Language

THE ASK

Join Us in Building the Future

1

Participate

Share your WG's key terms,
concepts, and pain points

2

Collaborate

Join our cross-WG sessions to
map connections

3

Champion

Be a liaison between KSWG
and your Working Group

Let's talk after the session

KSWG Team

THANK YOU



Let's talk after the session

KSWG Team